Network Traffic Analysis and Security Investigation Using Wireshark

Overview

This project documents a beginner-level investigation of network traffic using Wireshark on Windows 11. The goal is to understand common network protocols, observe DNS resolution and ICMP connectivity tests, inspect TCP communication, and document findings using packet-capture evidence.

Objectives

- Capture authorized network traffic.
- Identify source and destination IP addresses.
- Analyze DNS, ICMP, and TCP packets.
- Apply Wireshark display filters.
- Document observations, limitations, and troubleshooting recommendations.

Tools

- Wireshark
- Npcap
- Windows 11 Command Prompt
- Git and GitHub

Lab Activities

1. Record the local IP configuration.
2. Capture traffic generated by DNS lookup and ping tests.
3. Filter packets by protocol.
4. Inspect packet headers and responses.
5. Prepare screenshots and an investigation report.

Results

Results will be added after the lab tasks are performed and verified. No suspicious activity will be claimed without supporting evidence.

Ethics and Privacy

Only traffic that I own or am authorized to investigate is captured. Sensitive packet captures, credentials, and personal information are excluded from public uploads.

Project Status

In progress